ModelIt!

Student Activity Pack

Your Sneakers Are Made of Dinosaurs

How Scientists Engineer Materials That Don't Exist in Nature

Grade 6 | NGSS MS-PS1-3, MS-PS1-6 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Look at your clothes and belongings. Which items exist in nature and which were made by humans?**

2. **Where do you think plastic comes from originally?**

3. **Why might scientists create materials that don't exist in nature?**

4. **What makes a good material for a shoe sole? List the properties it needs.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Polymer Chain Length | Heat Treatment | Chemical Additives | Raw Material Quality

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Weak Plastic

Short chains, minimal treatment

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Super Strong

Maximum chain length, optimal heat treatment

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Flexible Film

Maximum plasticizer additives

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Cross-Linking Density — Chemical bonds between polymer chains that create a network — more cross-links mean stiffer, more heat-resistant material

Recycled Content — The percentage of material made from recycled sources rather than virgin petroleum — affects both properties and sustainability

Biodegradability — How quickly the material breaks down in the environment — a critical property for environmental sustainability

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If most of the stuff you wear, carry, and use every day doesn't exist in nature, where does it come from and what is it made of? Use evidence from your simulation to support your answers.

1 Why can the same base polymer become a rigid water bottle OR a flexible plastic bag?

2 What are the environmental trade-offs of synthetic materials — benefits vs. costs?

3 How is making a polymer similar to building with LEGO bricks?

4 Why is it important to develop plant-based alternatives to petroleum plastics?

5 How does understanding molecular structure help engineers design better materials?

8 STEM Challenge: Design a Sustainable Material of the Future

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YOUR ENGINEERING MISSION

Design a new material for a specific product that is both high-performance AND environmentally sustainable, replacing a petroleum-based plastic.

THE SCENARIO

A sportswear company wants to eliminate petroleum-based plastics from their products by 2030. Your materials engineering team must design a replacement material for running shoe soles that matches the performance of current materials while using sustainable raw materials.

Engineering Design Thinking

1. WHAT NATURAL OR RECYCLED RAW MATERIALS WILL YOU START WITH?
2. WHAT POLYMER CHAIN LENGTH AND HEAT TREATMENT WILL GIVE YOU THE NEEDED DURABILITY?
3. WHAT ADDITIVES WILL PROVIDE THE RIGHT COMBINATION OF CUSHIONING AND GRIP?
4. HOW WILL YOU MEASURE PERFORMANCE: BOUNCE, DURABILITY, WATER RESISTANCE, OR SOMETHING ELSE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Materials Engineers and Polymer Scientists develop new synthetic materials for products ranging from sneakers to spacecraft. They work on sustainability challenges and create materials that don't yet exist. They earn $70,000–$130,000/year.